



ENCE 307 ENGINEERING ECONOMICS

Er. Shrutee Karna

Department of Civil Engineering

COURSE CONTENT

I. Introduction

(2 hours)

- 1.1 Micro, macro and engineering economics (History, fundamental principle and application)
- 1.2 Terminology related to engineering economics
- 1.3 Economic decision and role of engineers in decision making
- 1.4 Cash flow and cash flow diagram

2. Market Economics

(3 hours)

- 2.1 Market, demand, supply and relationship
- 2.2 Elasticity, application of elasticity and government policies
- 2.3 Externality and market inefficiency
- 2.4 Market failure and firm behavior

COURSE CONTENT

3 Cost (8 hours)

3.1 Cost classification

3.1.1 Total, average, fixed, variable, and marginal costs

3.1.2 Direct, indirect, and standard costs

3.1.3 Cash versus book cost, manufacturing and non-manufacturing cost

3.1.4 Sunk cost, opportunity cost, element of cost, life-cycle cost

3.2 Cost estimation and control

4 Time Value of Money (6 hours)

4.1 Money (Type, functions and time value of money)

4.2 Simple and compound interests (Nominal, effective and continuous compounding)

4.3 Economic equivalence

4.4 Cash flow types (Single, uniform, linear gradient, geometric gradient and irregular)

COURSE CONTENT

5 Methods of Economic Analysis

(12 hours)

5.1 Capital budgeting

5.2 Minimum attractive rate of return (MARR)

5.3 Economic analysis of single and multiple projects

5.3.1 Payback period (Simple and discounted)

5.3.2 Equivalent worth (Net present, annual, future and capitalized)

5.3.3 Rate of return (Internal and external)

5.3.4 Public Sector economic analysis (Benefit cost analysis)

5.3.5 Financial and economic analysis

5.4 Weighted average cost of capital (WACC)

5.5 Repeatability assumption and co-Terminated assumptions

5.6 Multiple investment project alternatives (Dependent, independent and contingent)

COURSE CONTENT

6 Replacement Analysis

(5 hours)

- 6.1 Replacement strategies (Asset life and selection of challengers over defenders)
- 6.2 Economic service life of an asset
- 6.3 Replacement strategy for asset (Project with finite and infinite planning horizon)

7 Risk Analysis

(5 hours)

- 7.1 Origin of risk in projects
- 7.2 Risk analysis of projects (Sensitivity, breakeven and scenario analyses)
- 7.3 Decision tree

COURSE CONTENT

8 Depreciation and Taxes

(5 hours)

8.1 Concept and terminology

8.2 Depreciation calculation (Straight line, declining balance, sinking fund, sum of the year digit and modified accelerated cost recovery methods)

8.3 Tax and corporate income tax

8.4 Economic analysis (After-tax cash flow)

9 Measurement of Nation Income

(5 hours)

9.1 Gross domestic product (Components, real and nominal gross domestic product)

9.2 Unemployment (Measurement, job search, minimum wage law)

9.3 Inflation

9.3.1 Causes, effects and measurement of inflation

9.3.2 Constant and current cash flow

9.3.3 Equivalence calculation under inflation

9.3.4 Inflation controlling measures

9.4 Real and nominal exchange rates, fiscal budget and monetary policy

9.5 Financial statement

REFERENCES

1. Mankiw, N. G. (2017). Principles of economics (8th ed.). Cengage Learning.
2. Park C.S. (2016). Contemporary Engineering Economics: Prentice Hall, Inc.
3. McConnell, C. R., Brue, S. L., Flynn, S. M. (2020). Economics: Principles, problems, and policies (22nd ed.). McGraw-Hill Education.

EVALUATION

I. Internal Evaluation

[40 Marks]

- Class participation
- MCQ'S/Presentation
- Assignment
- Assessment I
- Assessment II

[4 marks]

[6 marks]

[10 marks]

[10 marks]

[10 marks]

2. Final Examination

[60 Marks]

EVALUATION

Chapter	Hours	Mark distribution*
1	2	3
2	3	5
3	2	3
4	6	8
5	12	16
6	5	6
7	5	6
8	5	8
9	5	5
Total	45	60

Chapter I

INTRODUCTION

Content

- I.1 Micro, macro and engineering economics (History, fundamental principle and application)
- I.2 Terminology related to engineering economics
- I.3 Economic decision and role of engineers in decision making
- I.4 Cash flow and cash flow diagram

Definition

- **Economics** is a social science that studies how individuals and society use limited (scarce) resources to satisfy unlimited human wants.
- The word Economics is derived from the Greek word Oikonomikos
Oikos = household
Nomos = law or management
- → Meaning “management of the household.”
- In ancient Greece, economics referred to efficient household management, where resources were carefully allocated to meet family needs.
- Xenophon, in his book Oeconomicus, explained economics as the art of managing household and agricultural resources.

Definition

- **Aristotle** distinguished between:
Oikonomia (proper use of wealth for welfare)
Chrematistics (unlimited wealth accumulation for profit)
- **Adam Smith**, known as the Father of Economics, defined “economics as the science of wealth,” focusing on production, division of labour, and national income.
- **Alfred Marshall** shifted the focus from wealth to people and defined economics as the study of mankind in the ordinary business of life, emphasizing human welfare.
- **A.C. Pigou** expanded welfare economics by stressing social welfare and the role of government in correcting economic inequalities and externalities.

Definition

- **Lionel Robbins** gave the most accepted definition, stating that economics studies human behavior as a relationship between ends and scarce means which have alternative uses, highlighting scarcity, choice, and opportunity cost.
- **Paul Samuelson** broadened the scope by including production, distribution, consumption, and economic growth over time, for present and future needs.
- **Modern economics** integrates all these ideas and focuses on efficient allocation of resources, economic growth, and improving living standards.
- Thus, economics today is concerned with wealth creation, human welfare, scarcity, choice, and sustainable development.

Definition

- **Engineering Economics** is a branch of economics that deals with the economic analysis of engineering projects and decisions. It helps engineers choose the best alternative among different technical solutions by comparing their costs and benefits over time.
- Engineering economics applies economic principles to engineering decision-making.
- It focuses on evaluating costs, revenues, savings, and benefits associated with engineering projects.
- Engineering economics is the application of economic principles to evaluate and select the most economical engineering alternative over its life cycle.

Definition

- The main objective is to ensure that a project is not only technically feasible but also financially economical.
- It considers the time value of money, meaning that money today is worth more than the same amount in the future.
- Engineering economics helps in selecting the best alternative among multiple options such as:
 - Different designs
 - Different materials
 - Different construction methods

Origin of engineering economics

Engineering economics emerged in the late 19th century as a response to the massive capital requirements and technical complexities of large-scale infrastructure projects, most notably the expansion of railroads. It evolved from early practical cost-benefit analyses into a formal academic discipline in the early 20th century.

I. Pioneering Work (Late 19th Century)

The formal origin of engineering interest in economic evaluation is traced to Arthur M. Wellington, a civil engineer.

The Railroad Influence: In 1877, Wellington published the first edition of “The Economic Theory of the Location of Railways”.

Key Contribution: He introduced the capitalized cost method to determine the optimal length and curvature of rail lines. His work shifted the engineer's role from merely building structures to recognizing that the "ultimate economy" is a primary concern in engineering decisions.

Origin of engineering economics

2. Development of the Modern Discipline (Early 20th Century)

- While Wellington laid the groundwork, the field was standardized through academic texts and theoretical bridges.
- The Bridge (1923): John Charles Lounsbury Fish published Engineering Economics: First Principles, which provided a critical link between early practical railroad economics and later formal textbooks.
- The "Father" of Engineering Economy (1930): Eugene L. Grant is widely considered the father of the discipline.
- He published the seminal textbook Principles of Engineering Economy in 1930.
- Grant introduced the concept of economic equivalence the idea that different sums of money at different times can be equal in value when considering interest rates.
- His work emphasized that business decisions should be based on prospective differences between alternatives and the time value of money.

Origin of engineering economics

3. Academic Formalization

Following the publication of Grant's work, the subject became an integral part of engineering curricula.

1942: The Industrial Engineering Division was formed within the Society for the Promotion of Engineering Education (S.P.E.E.), the forerunner of the American Society for Engineering Education (ASEE).

Key Early Figures: Other notable pioneers who organized the body of knowledge include H.G. Thuesen, E. Paul DeGarmo (who published Introduction to Engineering Economy in 1942), and Arthur Lesser, Jr., the founding editor of the journal The Engineering Economist.

Origin of engineering economics

4. Historical Context of Core Concepts

- Jules Dupuit (1840s): Though often categorized under general economics, this French civil engineer's work on the utility of public sector projects (e.g., bridges and roads) provided early foundations for cost-benefit analysis in engineering.
- Shift in Perspective: Modern engineering economics differs from the past by providing sound mathematical techniques to address uncertainty, risk, and resource conservation in public and private funding.

Micro, macro and engineering economics

Economics is mainly divided into two major branches:

- Microeconomics
- Macroeconomics

Micro-economics studies the economic behavior of individual units of the economy.

It focuses on:

- Individual consumers
- Individual firms
- Prices of individual goods and services
- Specific markets

Micro, macro and engineering economics

Key Areas in Microeconomics

- Demand and supply
- Elasticity of demand and supply
- Consumer behavior and utility
- Production and cost
- Market structures:
 - Perfect competition
 - Monopoly
 - Monopolistic competition
 - Oligopoly
- Factor pricing (wages, rent, interest, profit)
- Welfare economics

Micro, macro and engineering economics

- **Macro-economics** studies the economy as a whole. It focuses on aggregate indicators such as national income, inflation, unemployment, and economic growth.

Key Areas of Macro-economics

- National income (GDP, GNP)
- Inflation and deflation
- Unemployment
- Fiscal policy (taxation and government spending)
- Monetary policy (interest rates, money supply)
- Economic growth and development

Micro, macro and engineering economics

Microeconomics (Project / Individual Level)

- Cost comparison of cement, steel, RCC in a building project
- Deciding labour wages and number of workers on site
- Renting vs buying construction equipment
- Fixing tender price for one road or building
- Choosing survey method (Total Station, GPS, Drone) for a single site
- Pricing a land survey or mapping project

Micro, macro and engineering economics

Macroeconomics (National / Economy Level)

- Government budget for roads, bridges, hydropower
- Inflation affecting cement, steel, fuel prices
- Interest rates on construction loans
- GDP growth increasing infrastructure and housing demand
- National land policy, mapping programs, GIS projects
- Employment level in construction and geomatics sector

Terminology related to engineering economics

- **Interest:** The fee paid for the use of borrowed money (from the borrower's perspective) or the return earned on invested money (from the lender's perspective).
- **Interest Rate :** The percentage used to calculate the interest earned or paid over a specific time period (e.g., 5% per year).
- **Present Worth (PW) / Present Value (PV) :** The current value of a future sum of money or a series of cash flows, discounted at a specific interest rate.

Terminology related to engineering economics

- **Future Value (FV)** : The value of a current sum of money or a series of payments at some point in the future, compounded at a specific interest rate.
- **Annual Worth (AW) / Equivalent Uniform Annual Cost (EUAC)** : A uniform series of equal, end-of-period cash flows that is equivalent to a project's cash flows over its life, considering the time value of money.
- **Minimum Attractive Rate of Return (MARR)**: The lowest rate of return an investor or company is willing to accept for a project. It's often called the "hurdle rate."

Terminology related to engineering economics

Cash Flow & Evaluation Terms:

- Cash Flow

Movement of money into and out of a project over time.

- Cash Flow Diagram

Graphical representation of cash inflows and outflows versus time.

- Net Present Value (NPV)

Difference between present value of benefits and costs.

- Benefit–Cost Ratio (B/C)

Ratio of present worth of benefits to present worth of costs.

- Internal Rate of Return (IRR)

Interest rate at which NPV becomes zero.

Terminology related to engineering economics

Depreciation & Replacement Terms

Depreciation

Reduction in asset value over time due to wear, age, or obsolescence.

Salvage Value

Estimated value of an asset at the end of its useful life.

Book Value

Asset value after accounting for depreciation.

Useful Life

Period during which an asset is economically usable.

Replacement Analysis

Economic comparison between keeping old equipment and replacing it.

Terminology related to engineering economics

Cost

Monetary value of resources (materials, labor, equipment) used to produce a product or service.

Fixed Cost

Cost that does not change with the level of production (e.g., rent, insurance).

Variable Cost

Cost that varies directly with production output (e.g., materials, fuel).

Initial Cost (First Cost)

Capital required at the beginning of a project (purchase, installation).

Terminology related to engineering economics

Operating Cost

Expenses incurred during operation and maintenance of a system.

Sunk Cost

Past cost that cannot be recovered and should not affect future decisions.

Opportunity Cost

Benefit lost when one alternative is chosen over another.

Principles of Engineering Economics

1. **Develop alternatives:** Engineering decisions must be made by comparing two or more feasible options.
2. **Focus on differences:** Only the incremental costs and benefits between alternatives influence the decision.
3. **Consistent viewpoint:** Economic analysis should be performed from a single stakeholder's perspective.
4. **Common unit of measurement:** All costs and benefits should be expressed in monetary terms for fair comparison.
5. **Consider all relevant criteria:** Both quantitative and qualitative factors must be included in evaluation.

Principles of Engineering Economics

- 6. **Time value of money:** Money today is worth more than the same amount in the future due to interest.
- 7. **Proper interest rate:** The selected interest rate should reflect the cost of capital and risk.
- 8. **Risk and uncertainty:** Possible variations in costs and outcomes should be explicitly analyzed.
- 9. **Revisit decisions:** Economic decisions should be reviewed when new information or conditions arise.
- 10. **Benefit–cost relationship:** An alternative is economically acceptable only when its benefits justify its costs.

Economic decision and role of engineers in decision making

Economic Decision:

An economic decision is a choice made to use limited resources in the most effective way to achieve a desired objective. Since resources such as money, materials, time, labor, and equipment are scarce, every decision involves selecting one alternative and giving up another, which leads to opportunity cost.

Economic decision and role of engineers in decision making

Economic decisions involve:

- Comparing costs and benefits
- Evaluating different alternatives
- Considering the time value of money
- Assessing risk and uncertainty

Economic decision and role of engineers in decision making

Role of Engineers in Economic Decision Making

1. Identify Alternatives: Multiple designs, methods, and technologies.
2. Estimate Costs: Construction, operation, maintenance, and replacement costs.
3. Evaluate Options: Apply NPV, IRR, Benefit–Cost Ratio, Annual Worth.
4. Analyze Time Value of Money: Account for project life, inflation, and interest.
5. Allocate Resources: Optimize labor, materials, equipment, and time.
6. Assess Risks: Consider price fluctuations, delays, and technical uncertainties.

Economic decision and role of engineers in decision making

7. Plan Project Schedule: Determine cost-effective timelines.
8. Incorporate Sustainability: Environmental, safety, and social impacts.
9. Optimize Design: Balance technical feasibility and economic efficiency.
10. Support Decision-Making: Provide data and analysis for management or stakeholders.
11. Monitor and Control Costs: Track progress and adjust resources as needed.
12. Evaluate Alternatives for Maintenance or Replacement: Decide whether to repair or replace infrastructure.

Cash flow and cash flow diagram

Cash flow is the movement of money into (inflow) and out of (outflow) a project over time.

In engineering economics, it represents costs and benefits occurring at different time periods during a project's life.

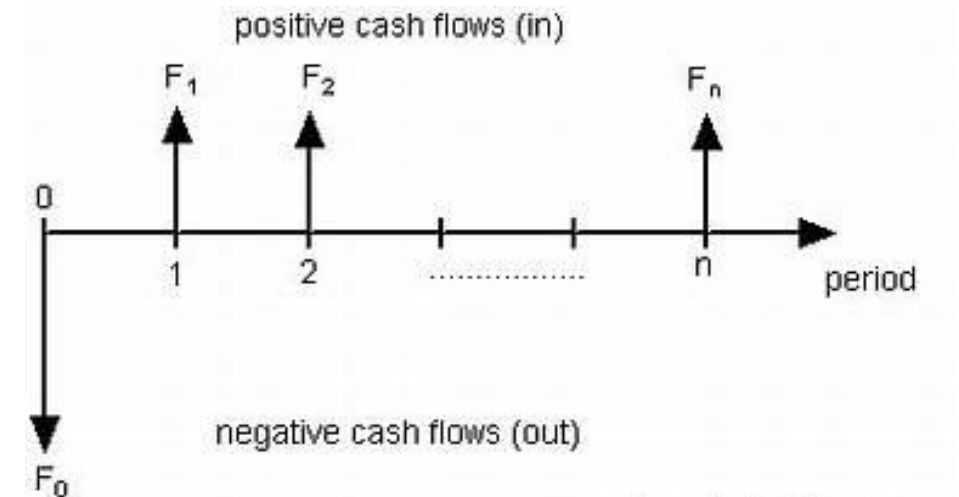
Cash Flow Diagram

A cash flow diagram is a time-based graphical representation of cash inflows and outflows, where:

Upward arrows ($\uparrow$) show cash inflow (+)

Downward arrows ($\downarrow$) show cash outflow (-)

The horizontal line represents time



Thank you for your attention!